COMP3004 Coursework 2 Q&A

1. **How can I install Ubuntu 18.04 or 20.04 Desktop VM?**

A: For the latest 20.04, you can download it from https://ubuntu.com/download/desktop

For previous version, e.g. 18.04, you can download it form: https://releases.ubuntu.com/18.04/

2. **How can I install/test mininet?**

A: After you have installed VM, you can install mininet in the following two approaches.

(1). Install mininet from source:

http://mininet.org/download/#option-2-native-installation-from-source

(2). Install mininet from packages

$sudo apt-get install mininet

http://mininet.org/download/#option-3-installation-from-packages

After mininet is installed, you can run '$sudo mn --test pingall' to test it.

3. **How can I install other packages?**

You can install e.g. Apache2 server, xterm, curl and openssh-server.

$sudo apt install apache2

$sudo apt install xterm

$sudo apt install curl

$sudo apt install openssh-server

4. **How do I know whether the Apache2 server is running and listening at port 80?**
$netstat -lnt (to check whether port 80 is listening)
$ps aux |grep apache* (to check whether apache2 server is running)

5. **How do I start/stop/restart Apache2 server?**
   $sudo service apache2 start/stop/restart

6. **How to install DASH player and segment the video**
   - Install x264 and gpac (sudo apt-get install x264; sudo apt-get install gpac)
   - Segmentation of video following Lab 8 on DASH to create .mpd file, and install dashjs under /var/www/html
   - If you have a working data segmentation/dashjs directory in one VM (e.g. following Lab 8 on DASH), you can copy all contents under /var/www/html directory to another VM under the same directory using a tarball (e.g., compress the files/directories under

/var/www/html into a .tar.gz file, then scp to the same location in another VM and decompress the file there).

- Details on how to use x264 and MP4Box for MPEG-DASH content generation, please see the following link: https://bitmovin.com/mp4box-dash-content-generation-x264/

7. How to install a web browser (e.g. google chrome) from command line

- $wget https://dl.google.com/linux/direct/google-chrome-stable_current_amd64.deb (to download google-chrome Debian package)
- $sudo dpkg -i google-chrome-stable_current_amd64.deb (to install google chrome Debian package)
- $google-chrome (to start google-chrome)

(Note: for ubuntu desktop version, you can also install google chrome from a web browser, e.g. firefox)

8. How to start Apache2 server on a virtual host of mininet?

- If apache2 is running under root system, stop it first under root system.
- Run 'xterm h1' under mininet prompt, then run 'apache2ctl -f /etc/apache2/apache2.conf' to start Apache2 server in h1 (ignore warning on server's fully qualified domain name). Note: if you run 'apache2ctl start' in h1, it will invoke 'systemctl start apache2', which will still start Apache2 in root system. To start Apache server in h1, you have to use command line command, i.e. run 'apache2ctl' with command line arguments.
- If you want to get rid of the warning message, you can add one line 'ServerName 127.0.0.1' at the end of apache2.conf file.
- Check whether the server is listening at port 80, initiate 'netstat -lnt' in h1.
- If you want to stop apache2 server on h1, just run 'apache2ctl stop' in h1.
- Find out the IP address of h1 (e.g. 10.0.0.1) by running 'ifconfig' in h1
- Now go to h2 (e.g. xterm h2), run 'curl 10.0.0.1'. curl should be successful.

9. How to start a web browser from another virtual host (e.g. h2)

- From h2, run 'google-chrome', it may complain about running google chrome as root.
- You can switch to your non-root user name (e.g. su your-VM-user-name) on h2.
- Then start google-chrome web browser in h2.
- If web browser starts correctly, you can enter 10.0.0.1/dashjs in URL (assume apache/dash server is started from h1). You can enter a proper *.mpd file for dash streaming (details see Lab 8 on DASH).
- You can test whether streaming is through mininet, by running, e.g., 'link s1 h1 down' under mininet prompt to see whether video streaming can be stopped (note: the video streaming should stop only after all buffered contents are played out). Before this test, make sure you have cleared all caches in web browser (in google chrome, go to settings, clear browsing data which includes all cached images and files).